Humans' Fast and Slow Responses to Competition from Internal Generative AI in Online Q&A Communities

Abstract

The advent of generative artificial intelligence (GenAI) has disrupted online question-and-answer (Q&A) communities and prompted debates on how platforms should respond. Our work examines a platform strategy of introducing platform-provided (internal) GenAI answers, an arrangement that differs fundamentally from the external general-purpose GenAI systems (like ChatGPT) studied in prior work: rather than operating privately and beyond a platform's authority, an internal answerer is deployed under the platform's own governance, displayed under an explicit machine-authorship label, and positioned alongside human contributors as an identified competitor. Taking a human-AI competition view, grounded in the two systems of thinking (fast-and-slow) theory, we argue that the total AI effect is the sum of two separable components: a fast response to the AI label and a slower response to the AI content. Using a randomized field experiment at a leading technical Q&A community, we show that the AI label invites human contribution because the community is skeptical of AI's capabilities, whereas longer AI content suppresses human contribution, and higher-quality AI content raises the quality of human answers as contributors reuse and build on it. Decomposing the total AI effect offers a theory-driven framework for reconciling seemingly conflicting findings in the literature and for anticipating human responses as AI capabilities advance. We contribute to the emerging literature on GenAI's impact on human society and provide practical insight for the adoption and application of GenAI on platforms.

Keywords: Generative AI, Human-AI Interactions, Online Q&A Communities, Two Systems of Thinking.

1. INTRODUCTION

For much of the internet's history, online question-and-answer (Q&A) communities served as a primary repository for human knowledge, where people posted problems and received answers that then remained available to everyone who followed. Generative artificial intelligence (GenAI) has unsettled that role. Several recent studies document a sustained decline in user participation on such platforms following the release of ChatGPT (Burtch et al. 2024; Quinn & Gutt 2025; del Rio-Chanona et al. 2024; Sanatizadeh et al. 2025; Xue et al. 2026). These results raise a pressing question about whether such communities remain viable, and about what, if anything, a platform can do in response.

A platform cannot stop users from consulting external GenAI tools such as ChatGPT, nor recover the questions those tools answer privately before they reach the community (Xue et al. 2026). One response, however, lies fully within a platform's control, that is, deploying its own large language model (LLM) to answer posted questions, an arrangement we term "internal GenAI." As Figure 1 illustrates, a newly posted question triggers a labeled AI answer that human contributors encounter as they decide whether to respond. Internal GenAI is not simply external GenAI relocated inside the platform. It is a distinct sociotechnical and governance arrangement. Where an external tool operates outside the community and beyond the platform's authority, an internal answerer is deployed under the platform's own governance, displayed under an explicit machine-authorship label, and positioned in the answer thread alongside human contributors. These features change the terms on which humans encounter the AI. The platform, rather than an outside vendor, decides whether an AI answer appears, how it is presented, and how capable it is, so its presence becomes a design choice rather than an environmental shift. The explicit label carries an institutional signal about who produced the answer and reshapes how the community assesses its legitimacy and reliability. And because the AI now competes for the same readers as any human answer, it enters the community as an identified competitor rather than a private substitute. Since many leading platforms, including Quora, have already deployed internal GenAI, understanding how this arrangement affects human contributors is a first-order design question. A further benefit of this setting, discussed below, is that the visible label and content of the AI answer make its influence directly observable, which allows us to examine effects that are hidden when GenAI is consulted privately.

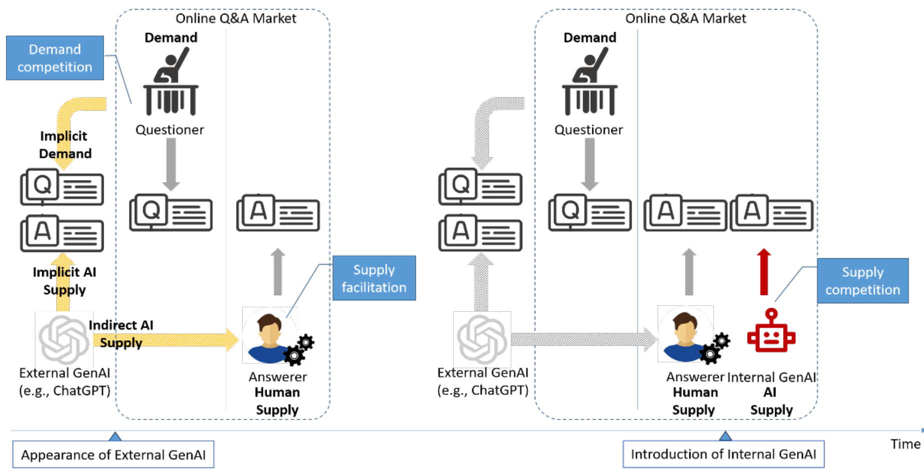


Figure 1. Difference between external GenAI and internal GenAI in online Q&A communities

Whether an internal AI answer helps or harms an online Q&A community is not obvious, because the logic behind most prior work predicts one outcome while the one comparable study finds the opposite. Most prior studies examine external GenAI and rest, implicitly, on a substitution logic in which the AI privately does work a human would otherwise have done, so question supply declines (Burtch et al. 2024; Quinn & Gutt 2025; del Rio-Chanona et al. 2024; Sanatizadeh et al. 2025; Xue et al. 2026). This logic fits the aggregate declines observed after ChatGPT's release, but it was built on external GenAI and can predict only decline. Su et al. (2025), by contrast, study an internal GenAI answerer, the same type of intervention we examine, and find that human contributions rose in both number and length. The prevailing logic and the one internal precedent thus disagree, and neither provides a framework for when each outcome should occur.

Therefore, we ask how a labeled, platform-provided AI answer affects human contribution. We separate this into two concrete decisions. First, does a contributor choose to answer a question the AI has already addressed? Second, once a user decides to answer, how good is the answer they produce?

Our central argument is that the effect of an AI answer cannot be summarized by a single quantity, because the presence of an AI answer is not a single condition. We adopt a competition lens (Chen et al. 2007; Chen 2009), under which contributors treat the AI answer as any competing contribution, and draw on the distinction between fast and slow thinking (Evans & Stanovich 2013; Kahneman 2003, 2011) to explain why different features of that competitor are read differently and bear on different decisions. We distinguish two features. The AI label is the visible marker of machine authorship; it does not vary in degree and carries whatever a community already believes about AI. The AI content is what the AI actually wrote, captured by its length and quality, and varies across answers. In a community that regards AI-generated technical answers with skepticism, and such skepticism has an empirical basis given that a majority of ChatGPT answers to programming questions have been found to contain incorrect information (Kabir et al. 2024), these features need not push the same way. The label can invite a human to step in, the length can crowd that human out, and the quality can give a contributor something to build on. The net effect depends on how these features combine.

We test this account using a randomized field experiment at SegmentFault, a leading Chinese technical Q&A community. Beginning in September 2023, the platform randomly assigned each newly posted question either a labeled internal GenAI answer or none. Because external GenAI remained available for private use in both conditions and continued to divert demand from the community, our estimates capture the incremental effect of adding internal GenAI rather than the effect of GenAI relative to an AI-free environment. Two features make the setting well suited to our question. First, the AI answer is explicitly labeled, so the AI label is observable to both community members and us, unlike external GenAI consulted privately. Second, because the label is fixed while content varies across questions, the design allows us to trace how human contribution moves with the measurable characteristics of the AI answer.

Our results show that the AI label and the AI content move human contribution in opposing directions, and that they act on different decisions. On entry, human answer supply rises with the presence of the AI label, consistent with contributors treating a labeled machine answer as a weak competitor worth answering over, while a longer AI answer lowers supply. On crafting, a higher-quality AI answer is associated with higher-quality first human answers, consistent with contributors reusing the AI answer's sound material and following the direction it establishes rather than starting from a blank page. This pattern follows directly from our account: the feature registered at a glance, the label, governs whether a person enters, while the feature apparent only on closer reading, quality, governs how good the answer they produce becomes. The absence of a supply response to quality is consistent with quality being assessed only after a contributor has already committed to answering. Because AI answers in our setting typically carry substantial content, the length relationship dominates the label on supply, so the average effect on supply is negative even as the label alone works in the opposite direction.

This paper makes three contributions. First, we distinguish the roles of the AI label and the AI content in shaping human contribution, which the external-tool literature cannot do because there the AI is either an invisible drafting aid or a private substitute that never appears inside the community as an identified author. Because the label is fixed and the content varies, we recover the role of the label by tracing how contribution responds to the AI answer's content and characterizing what remains as content diminishes, an approach the labeled deployment makes available to us. Second, because our estimates tie the effect to the measurable characteristics of the AI answer rather than to a fixed average, the account is forward-looking; as AI capability advances and answers grow longer and better, one can ask how human contribution should respond, while recognizing that the label's role is contingent on community beliefs that may themselves change. Third, for practice, we show that platforms can manage the AI label and the amount of AI content as two independent choices. Regardless of the precise beliefs that drive the label's effect, we recommend that platforms manage the visible extent of AI content and the labeling of AI authorship as separate decisions, because a platform can improve a capable internal GenAI to raise human answer quality without necessarily discouraging participation.

The remainder of the paper proceeds as follows. Section 2 reviews the literature on GenAI in online Q&A communities and on human responses to prior and machine-generated contributions. Section 3 develops the hypotheses. Section 4 describes the research context. Section 5 presents the empirical strategy, Section 6 reports the results and mechanism tests, and Section 7 discusses the findings, their limitations, and their implications.

2. LITERATURE REVIEW

Our study draws on three literatures. The first concerns how GenAI has reshaped online Q&A communities; we organize this literature by the structural role AI plays, because that role determines both what each study is able to identify and what it necessarily leaves open. The second concerns what drives a community answerer's decision to answer a question that already has prior contributions. The third, on human responses to identified machine agents, bridges the first two.

Table 1. Summary of empirical research on the effects of generative AI on online Q&A communities

Study	Platform (domain)	Source of variation	Unit of analysis	AI identifiable	Question supply	Answer supply
Role 1. External GenAI as a platform competitor						
Xue et al. (2026, ISR)	Stack Overflow (technical)	ChatGPT release	Topic-day	No	–; remaining questions higher quality	
Quinn & Gutt (2025, JMIS)	Stack Exchange (technical)	ChatGPT release	Topic-week	No	–	
Sanatizadeh et al. (2025, I&M)	Stack Exchange (technical)	ChatGPT release	Post-day	No	–; remaining questions more complex	
Rao et al. (2026)	Stack Overflow (technical)	ChatGPT release	User-week	No	+ (individual level); remaining questions longer & more novel	
Burtch et al. (2024)	Stack Overflow (technical)	ChatGPT release	Topic-week	No	–	
del Rio-Chanona et al. (2024)	Stack Overflow (technical)	ChatGPT release	Topic-week	No	–	
Wang & Zheng (2024)	Stack Exchange (technical)	ChatGPT **ban**	Topic-week	No	+	
Role 2. External GenAI as the answerer's instrument						
Shan & Qiu (2026, ISR)	Stack Overflow (technical)	ChatGPT release	User-day	No		+; shorter, more readable
Fang & Zhou (2026, I&M)	Zhihu (general)	ChatGPT release	User-month	No		+; shorter
Sanatizadeh et al. (2025, I&M)	Stack Exchange (technical)	ChatGPT release	Post-day	No		–; longer, less readable
Borwankar et al. (2026)	Stack Overflow (technical)	ChatGPT **ban**	Answer-user-day	No		Answers shorter & more negative
Li & Kim (2025)	Stack Overflow (technical)	ChatGPT release	User-day	No		–
Wang & Zheng (2024)	Stack Exchange (technical)	ChatGPT **ban**	Topic-week	No		+; acceptance +
Role 3. Internal AI as the answerer's competitor						
Su et al. (2025)	Quora (general)	Internal GenAI (applied on old questions)	Question-week	Yes		+
This study	SegmentFault (technical)	Internal GenAI (applied on new questions)	Question	Yes	Not applicable (due to question level randomization)	–

Notes. 1. Signs ("+"/"–") denote the direction of each study's estimated effect; blank cells indicate outcomes not reported here by design.
2. Sign convention for ban-based studies. Most studies identify effects from the introduction of ChatGPT, so their signs directly capture the effect of AI becoming available. Two studies instead exploit a ChatGPT ban (Wang & Zheng 2024; Borwankar et al. 2026). Because a ban is the withdrawal of AI, the raw estimates in these studies measure the effect of removing AI and are the mirror image of the release-based estimates. To keep all rows comparable, we reverse their signs. For example, Wang & Zheng (2024) find that banning ChatGPT decreases question supply, answer supply, and answer acceptance; equivalently, the availability of ChatGPT increases question and answer supply.
3. Su et al. (2025). This is an internal-GenAI setting on Quora. The questions were created before the observation panel began, and platform AI answers were introduced only at the policy launch (≈ Jan 2023). Following their design, the sample keeps "only those questions that were created before the start of our panel data collection (i.e., November 2022)," and treated pages receive AI answers when "AI answers are presented on treated question pages" at launch. The AI answer is introduced long after the original question was posted (average pre-policy tenure is roughly 49 months).

2.1. The Impact of GenAI on Online Q&A Communities

A large and fast-growing literature documents the consequences of GenAI for online Q&A communities. Table 1 summarizes this work, grouped by the structural role of AI. As illustrated in Figure 1 and Table 1, the review yields three roles. AI can answer incoming questions and compete with the platform itself; AI can serve as the answerer's instrument; or it can enter as an identified rival contributor. The first two roles have been studied entirely in the context of external GenAI, tools that operate independently of any platform. The third role, which our study examines, is occupied by internal GenAI, which constitutes a distinct phenomenon from external GenAI.

2.1.1. External GenAI as a Platform Competitor

The first and largest stream examines AI operating *before* questions reach the community. General-purpose tools such as ChatGPT answer questions directly, and the consistent finding across platforms and time windows is a decline in question supply (Burtch et al. 2024; Quinn & Gutt 2025; del Rio-Chanona et al. 2024; Sanatizadeh et al. 2025; Xue et al. 2026). A notable exception is Rao et al. (2026), who find that users who actively employed ChatGPT to generate answers subsequently posted more questions. These questions were also longer and more novel, and attracted more upvotes. Consistent with Rao et al. (2026), Wang and Zheng (2024) exploit the ChatGPT ban and find that restricting access to ChatGPT decreased question supply. This pattern suggests a complementary effect at the individual user level, one that contrasts with the aggregate decline documented in studies examining platform-wide trends. The overall reduction in question volume results primarily from the fact that simpler questions are now answered by ChatGPT. Consequently, the remaining questions tend to be more complex and higher-quality (Quinn & Gutt 2025; Sanatizadeh et al. 2025; Xue et al. 2026). In this stream, AI is a competitor to the *platform*: it competes for the question itself and does not appear inside the community.

This stream establishes an important empirical baseline, and Xue et al. (2026) show that the decline is sustained rather than transitory. By construction, however, it speaks to whether questions arrive, not to how answerers behave once a question has arrived. It also describes an environment that platforms cannot control, since users cannot be prevented from consulting external tools. The managerially tractable question, that is, what a platform should do about the questions that do arrive, lies outside its scope.

2.1.2. External GenAI as the Answerer's Instrument

A second stream examines AI as an *input* to human answers. Here AI lowers the cost of drafting a response, and the human retains authorship. The relationship is collaborative rather than competitive, that is , human and AI work together in a single answering process. Several studies find that this raises answer supply (Fang & Zhou 2026; Shan & Qiu 2026), and evidence from the reverse direction points the same way, as bans on AI-generated content reduce both answer volume and acceptance rates (Wang & Zheng 2024). Other studies report declines in qualified contributions (Li & Kim 2025; Sanatizadeh et al. 2025), so the evidence is not uniform. Findings on answer style are similarly divided, with Sanatizadeh et al. (2025) reporting longer and less readable answers and others shorter and more readable ones (Borwankar et al. 2026; Fang & Zhou 2026; Shan & Qiu 2026).

Part of this divergence might reflect differences in platform, unit of analysis, and how answer style is operationalized. Taken together, these studies establish an important and non-obvious result: AI adoption changes not only how much answerers write but how they write. This stream cannot pin down how AI produces these effects, however, because AI's contribution is unobservable. A human answerer's private use of AI cannot be observed by other community members, and in most cases cannot be observed by the researcher either. Therefore, studies here can detect that answering behavior changed, but cannot measure how much of the answer the AI produced or connect that contribution to the observed change.

2.1.3. Internal GenAI as the Answerer's Competitor

A third and smaller stream examines GenAI that posts answers under its own explicit label, alongside human contributors, on the same questions. Here GenAI acts as a competitor to the human answerer, because it contends for the same viewers' attention, and both the source and the content of its contribution are visible. To date, only Su et al. (2025) have examined this configuration empirically. Studying Quora's deployment of its Sage bot, they report positive effects on human contribution, including increases in both the number and the length of answers. Two features of their setting, however, differ from a genuinely competitive configuration in ways that plausibly shape the nature of the effect.

The first is *timing*. The AI answer is introduced long after the original question was posted (average pre-policy tenure is roughly 49 months), by which point at least one human answer is already present (Su et al. 2025, pp. 15, 22). AI thus enters as a retrospective supplement to an established discussion rather than as a competitor that human contributors confront on arrival. The second is domain. Although Quora hosts some technical forums, it is a typical general-interest platform spanning topics such as philosophy, personal advice, and cultural discussion, where the value of a human contribution rests substantially on subjective interpretation and lived experience that AI cannot easily replace. In technical communities, by contrast, contributions are evaluated primarily on correctness, so competition between an AI answer and human answers may be considerably stronger. The existing literature does not fully develop this possibility, and we investigate it here.

2.2. How Content and Label Shape Human Contribution

2.2.1. Human Responses to Prior Contributions in Online Q&A Communities

A broad body of work identifies determinants of participation. These include question formulation and formatting (Burke et al. 2010; Calefato et al. 2018; Li et al. 2023), monetary and non-monetary incentives (Berger et al. 2016; Jiang et al. 2024; Wang et al. 2022), feedback and reciprocity (Li et al. 2020; Moon & Sproull 2008), and gamification (Chen et al. 2022; Meng et al. 2013). Because internal GenAI acts as an answerer, we focus on the dynamics of answering rather than on this broader set of determinants. In particular, when a potential contributor arrives at a question, that question usually already displays some prior answers, which shape the response of subsequent human answerers.

Prior work suggests that existing content shapes subsequent answering through four features: presence, volume, quality, and source. The first two features concern whether a new answer is still needed. The mere presence of prior answers can reduce a potential contributor's perceived marginal value, because the question appears at least partially solved; this logic is consistent with classic free-riding arguments, in which individual incentives decline as others have already contributed (Olson Jr 1971; Zhang & Zhu 2011). Beyond presence, the amount of existing content further affects the perceived room for contribution. As answers accumulate, each additional answer tends to add less new information, and threads with more prior answers may therefore offer fewer opportunities for a newcomer to make a distinctive contribution (Zhang et al. 2022). The third feature concerns the evaluative standard created by prior content. High-quality answers signal the level of effort and expertise expected in the thread, thereby shaping the quality of subsequent contributions (Shi et al. 2024).

Finally, the source of a prior answer provides a social cue about who has already spoken. A contributor's identity or standing can affect how newcomers assess their own relative position and whether their contribution is likely to be valued; consistent with this comparison logic, Chen et al. (2010) show that information about one's standing relative to peers changes subsequent contribution behavior. Together, this literature shows that prior contributions matter not only because they provide information, but also because they alter the perceived need, opportunity, benchmark, and social meaning of adding another answer.

Relevant to the source effect is the labeling of a prior contribution as human or AI. Because across this literature the prior contributor is another human, it leaves open whether presence, volume, and quality carry the same meaning when that contributor is identified as an AI.

2.2.2. Human Responses to Identified Machine Agents

Because internal GenAI is explicitly labeled, our setting activates a literature on how people respond to machine agents they know to be machines. This work establishes that disclosure alone changes behavior. People discount algorithmic advice after observing it err, even when the algorithm outperforms humans (Dietvorst et al. 2015), while over-weighting algorithmic judgment relative to human judgment in other conditions (Logg et al. 2019). The direction of the effect depends on how subjective or human-dependent the task is perceived to be (Castelo et al. 2019), which implies that the meaning of an AI label is domain-specific rather than fixed. Related literature shows that responses to capable machines are not confined to substitution or withdrawal: exposure to AI has been shown to increase the novelty of subsequent human decisions (Shin et al. 2023), and studies of AI-exposed labor markets document both displacement and adaptive repositioning toward tasks where human value is distinctive (Demirci et al. 2024; Hui et al. 2024; Yiu et al. 2026).

Two implications follow for online Q&A communities. First, an AI label may carry motivational content in its own right, independent of what the AI actually wrote. Second, a capable AI competitor may induce humans' motivation to outperform, and it may raise rather than lower the quality of human contributions. Whether these implications hold in online Q&A communities remains to be tested, which is the task we take up.

2.3. Positioning and Contribution of Our Study

As reviewed in Section 2.1, this study focuses on competition between internal GenAI and human answerers, whereas prior work on external GenAI examines competition with the platform and the invisible facilitation of human answers. To clarify how our study differs, Table A in the Appendix positions it against the three most closely related studies along the dimensions that drive our identification. Four features of our setting make our study distinctive. First, the GenAI is deployed by the platform under direct governance. Therefore, its presence is a design choice rather than an uncontrollable environmental shift, as in Xue et al. (2026) and Shan & Qiu (2026), which places the findings within managerial reach. Second, the GenAI responds almost immediately, before any human contributor arrives, so human contributors face the most competitively demanding configuration, unlike in Su et al. (2025). Third, the AI responses are explicitly labeled, so we can estimate the effect of the AI label separately from content attributes such as length and quality. Fourth, the assignment of AI answers is randomized at the question level, which permits clean causal inference and lets us compare questions whose earliest answer is AI-generated with otherwise identical questions whose earliest answer is human-generated.

Our study therefore addresses a question central to platform design and community management: how platform-deployed internal GenAI, layered on top of existing external GenAI availability, affects human contribution in an online Q&A community.

3. THEORETICAL DEVELOPMENT

3.1. Two Decisions in a Competitive Encounter

Consider the following situation, which mirrors the domain we examine. A member of a technical Q&A platform posts a question seeking a solution to a concrete problem. Shortly afterward, and before any human has responded, the platform attaches an answer generated by its own GenAI system. A human answerer who arrives at the thread later sees that a machine has already responded and must decide two things: whether to answer at all, and, if so, how to construct that answer.

We restate this as two concrete decisions and use the labels throughout. First, the answerer makes an "entry" decision, a yes-or-no choice about whether to contribute an answer to a question the AI has already addressed. Second, if they choose to enter, the answerer makes a "crafting" decision, which concerns both how good the answer will be and how it will be produced. Entry can be settled quickly, before the AI answer has been read closely; the crafting decision is a matter of degree and is where the answerer's time and knowledge are actually committed.

Two features of the encounter organize the theory that follows. First, the AI answer is explicitly labeled, so its source is apparent at a glance. Second, the length of the answer is likewise apparent before its substance is read. Thus the answerer can register both who produced the existing answer and how much has been written before engaging with what it actually says. Its quality, by contrast, becomes available only through the effort of reading.

We treat this encounter as a competitive episode. We adopt a competition lens because it matches the situation the platform has created: the AI answer sits in the same thread and on the same screen as any human answer that follows, so a later human answer must be worth reading in the presence of the AI answer. The competition lens also generates richer predictions than a substitution account. Prior work treating generative AI as a substitute for human labor establishes something important: these systems can perform much of the routine work human contributors once did (Burtch et al. 2024; Quinn & Gutt 2025; del Rio-Chanona et al. 2024; Sanatizadeh et al. 2025; Xue et al. 2026), but that account predicts only decline, because if the AI does the work the answerer would otherwise have done, the answerer has less left to do and writes less. It offers no route by which an AI answer could lead an answerer to write more, or to write better. We argue that a competitive framing provides exactly those routes.

3.2. The AI Answer as a Profile of Three Signals

Research on competitive interaction holds that how an actor responds to a competitor depends on how the actor appraises that competitor, and that this appraisal is built from observable features, including who the competitor is, how capable it appears, and how much effort it appears to have invested (Chen et al. 2007; Chen 2009). It follows that the effect of an AI answer cannot be summarized by a single quantity. The AI answer must instead be read as a profile of signals.

We take the AI answer to carry three signals. The source, which we call the "AI label," tells the answerer what kind of competitor this is (i.e., machine authorship is present). The "AI content" is what the AI actually wrote, and we decompose it into two measurable features: length signals how much of the answerable ground the competitor appears to have taken, and quality signals how good that content is (e.g., whether its reasoning is sound and its solution reliable). The AI label does not vary in degree, so it produces a level effect. Length and quality vary continuously from one answer to the next, so they produce slopes that determine how the consequence of an AI answer changes as its content changes. Our central claim, then, is that as

AI answers grow longer, and as they grow better, the consequence of having one under a question changes in ways that a single average cannot capture.

3.3. Linking Signals to Decisions Through Fast and Slow Thinking

The three signals do not all arrive at the same moment relative to the two decisions, and this timing is what ties each signal to a particular decision. Source and length are "glance" signals. The AI label is plainly visible, and so is the physical size of the answer, and neither requires reading. Quality is different in kind. An answerer learns how good an AI answer is only by reading it and checking its reasoning or its code against the specific problem the questioner posed. To connect signals to decisions, we draw on the two systems of thinking account (Evans & Stanovich 2013; Kahneman 2003, 2011).

In that account, System 1 operates automatically, demands almost no effort, and responds to surface cues, whereas System 2 is deliberate and effortful and engages only when the situation appears to justify the cost (Kahneman 2011). The two systems classify both the signals and the decisions. Among the signals, the AI label and length are surface cues available at a glance and are thus processed by System 1, while quality can be established only through the deliberate evaluation characteristic of System 2. Among the decisions, entry is settled quickly and before close reading, so it is a System 1 decision, whereas the crafting decision commits the answerer's time and knowledge and is thus a System 2 decision.

This classification yields a single organizing principle, which we label the "same-level" principle: a signal moves a decision when the two occupy the same level of processing, and cross-level effects are null. To state this another way, a glance signal bears on the glance decision, and a reading signal bears on the reading decision, but a signal processed at one level does not systematically shape a decision made at the other. Note that this principle commits us in advance to specific null predictions. A framework that predicted an effect in every combination of signal and decision could not be wrong in any interesting way, so the null cells matter as much as the others. They are tests the account must pass. The remainder of this section develops six hypotheses from this principle, which are arranged in Figure 2.

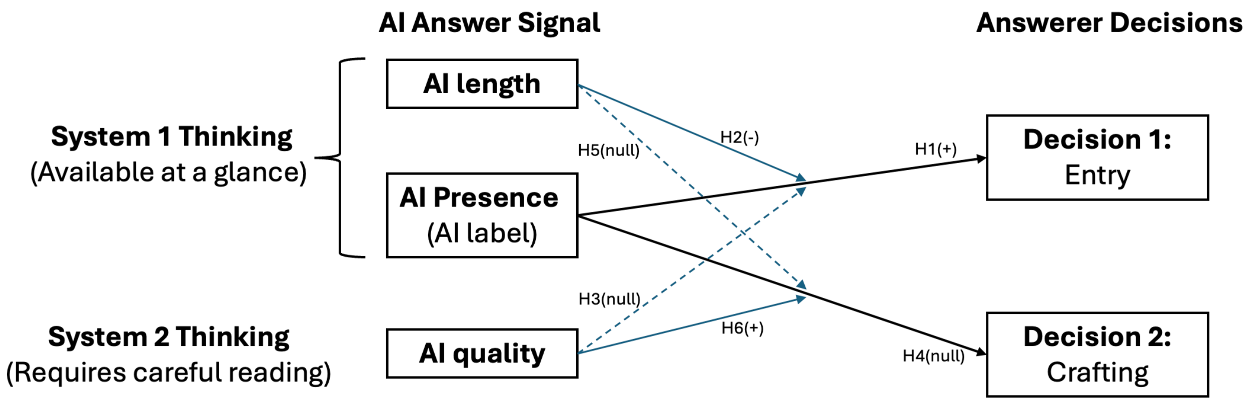


Figure 2. Conceptual framework

3.4. Hypotheses on The Entry Decision

3.4.1. Label and Length Shape Entry (System 1)

The AI label can shape entry because it is available before any content is read. We argue that the label leads the answerer to appraise the AI, and that a "weak-competitor" appraisal strengthens the incentive to enter.

Noticing a simple visual marker is enough to know that the answer came from the platform's AI system. System 1 then judges the competitor's capability through the availability heuristic, so the judgment rests on whatever beliefs about machine-generated answers come to mind most readily (Tversky & Kahneman 1973). AI's capability may change over time, but in the early period of GenAI during which our study was conducted, two beliefs are relevant.

The first belief is a general impression that generative text is bland and formulaic, a quality that popular discussion captures with the phrase "AI slop," which means the output follows a predictable template rather than saying anything sharp or particular. The second belief is specific to technical Q&A, and it is the one our argument rests on. Members of technical communities were openly skeptical of AI-generated answers when GenAI appeared, and their skepticism has an empirical basis. Kabir et al. (2024) report that more than half of ChatGPT's answers to Stack Overflow questions contain incorrect information, that these answers are considerably more verbose than human answers, and that readers miss the errors in a substantial share of cases, which is why Stack Overflow banned AI-generated answers. Disclosure compounds the effect, since information is judged less accurate when it is identified as machine-generated (Longoni et al. 2022). The consequence is not just doubt but cost: an answer that looks plausible but may be wrong has to be verified before it can be used, and verifying it means working through the reasoning oneself. An AI answer therefore does not close a question the way a trusted human answer would.

Appraising the AI as a weak competitor pushes toward entry. The AI answer sits under the same question, faces the same readers, and competes for the same attention, which is precisely the condition under which an actor is most likely to notice a competitor and respond (Chen et al. 2007; Chen 2009). A weak competitor makes responding attractive, because the risk of losing is low and a competent developer can provide the correct answer the AI failed to provide.

H1. The AI label alone increases human answer supply.

The second glance signal is length, and it pushes entry in the opposite direction. Length is processed through attribute substitution, in which System 1 replaces a difficult judgment with an easier one associated with it (Kahneman 2011; Kahneman & Frederick 2002). The difficult judgment is how much of the question the competitor has covered; the easy substitute is the answer's physical size. Our setting makes this substitution reliable, because on SegmentFault technical answers are broken into code blocks, numbered steps, configuration snippets, and shell commands, so a long answer in this format appears without reading to enumerate several possible causes or several fixes, and length becomes a credible indication of how much ground has been taken. What matters at entry is how much room is left. Thus the more of the question the existing answer appears to cover, the less room appears to remain, and the weaker the incentive to enter.

H2. The effect of an AI answer on human answer supply becomes more negative as the length of the AI answer increases.

Two points deserve emphasis. First, this is a claim about coverage, not quality, because length is far easier to perceive at a glance than quality, and an answer can be comprehensive and wrong at the same time (Kabir et al. 2024). Second, H2 complements H1 in describing the overall effect of a glance. The AI label's contribution is relatively stable, while length varies and shifts the perceived room for a contribution. This is why the consequence of adding an AI answer cannot be a single number. It is positive where the AI content is thin and negative where it is thick, and the point at which it changes sign depends on the content a given AI answer provides and, over time, on the continuing improvement of AI.

3.4.2. AI Answer Quality Does Not Shape Entry (System 2)

Quality is a System 2 signal, and entry is a System 1 decision, so the same-level principle predicts no systematic effect of AI answer quality on entry. Quality becomes available only afterward, because it requires reading the answer through and checking it against the specific problem. Entry is therefore shaped mainly by what arrives first. This argument rests on the division of labor between the two systems. System 1 forms an appraisal from the cues at hand and returns a disposition to act or to pass; System 2 is then in a position to examine that disposition and either endorse it or overturn it. The two-systems account emphasizes that System

2 normally endorses, because it is recruited only when the fast route fails to deliver a confident answer, and not recruited when the fast route delivers one cheaply, which is the law of least effort at work (Kahneman 2011). Here the fast route does deliver a confident answer. The label and the length together produce a coherent impression of how much ground the competitor has taken, so nothing calls for the more expensive judgment that assessing quality would require.

Even an answerer who does read the AI answer tends to confirm the first impression. An answer looks comprehensive because of how it is laid out, but correctness depends on whether the code runs, whether the named libraries and versions are the ones actually in use, and whether the diagnosis fits the setup described. Checking these things means working through the reasoning, which is demanding even for skilled programmers (Lee & Lee 2025), and errors in AI answers often go undetected for exactly this reason (Kabir et al. 2024). Careful reading is thus costly to start and unreliable once started, so it overturns the first impression less often than one might expect.

Reversals do occur in both directions. An answerer who meant to pass may find the AI answer mistaken and write up their own answer after all; an answerer who meant to contribute may find the AI answer sound and stop. We do not claim that either pattern is impossible. We claim that neither is systematic enough to produce a detectable relationship between the quality of an AI answer and the rate at which human answers appear.

H3 (null). The effect of an AI answer on human answer supply does not vary appreciably with the quality of the AI answer.

3.5. Hypotheses on the Crafting Decision

3.5.1. Label and Length Do Not Shape Human Answer Quality (System 1)

The crafting decision is made after the answerer has committed to answering and has thus read the AI answer through. Because the AI label and length are glance signals processed by System 1, while the crafting decision is the deliberate System 2 judgment of how to construct the answer, the same-level principle predicts that neither glance signal systematically shapes human answer quality. Once the answerer is engaged in evaluating the content itself, the surface cues that governed entry are superseded by what reading reveals.

Consider the AI label first. At entry, the label carried motivational content, because it identified the competitor as a machine and, in this skeptical technical community, invited a response (H1). That role is exhausted at the entry margin. When the answerer turns to constructing the answer, the answer is built from what the AI content actually contains, rather than from the bare fact of machine authorship. It could be argued that the weak-competitor appraisal formed at a glance persists and holds effort down, and we take this objection seriously. However, the two-systems account predicts that once System 2 is engaged, the crafting of the answer tracks the concrete content the answerer reads rather than the abstract source cue. We therefore predict no systematic label effect on the crafting decision.

H4 (null). The AI label alone does not affect the quality of the human answer.

Length is superseded for the same reason, reinforced by the beliefs of this population. Once the answerer has read the AI answer, its substance is directly available, so there is no reason to consult length as a proxy when the content itself is available to anchor on and build from. More specifically, AI answers to technical questions are substantially longer than human answers while being incorrect in more than half of cases (Kabir et al. 2024), and an answerer who holds this belief treats length as an indication of how much has been attempted rather than how well it has been done. Deliberation therefore discounts the cue rather than defers to it. We would not claim that length plays no role at all in the crafting decision, since an answerer may retain some residual impression of the competitor's scope from the initial scan. We claim only that this residue is too weak to generate a systematic relationship between the length of an AI answer and the quality of the human answers that follow.

H5 (null). The effect of an AI answer on human answer quality does not vary with the length of the AI answer.

3.5.2. AI Answer Quality Shapes Human Answer Quality (System 2)

Quality is a System 2 signal and the crafting decision a System 2 decision, so the same-level principle predicts a genuine effect. An answerer who has decided to contribute must still decide how good the answer will be and how to produce it. Because questions on technical Q&A platforms rarely state what would count as an adequate reply, the goal is underspecified. Having read the AI answer closely, however, the answerer now holds a concrete example of a reply to this specific question, one that shows how it should address the problem, whether it should explain the cause or simply supply the fix, and what a working solution looks like.

How useful this is depends on how good the AI answer is. A weak AI answer (e.g., a shaky diagnosis or code that may not run) offers little the answerer can trust or reuse, so the answerer must construct an answer largely from scratch. A strong AI answer both clarifies what a good answer looks like and provides something to build on. That support can take more than one form. Sometimes the AI answer contains sound material that can be adapted, so the answerer reuses a correct diagnosis or working code rather than reproducing it. Other times it contributes less finished content but still points toward the right approach, such as identifying the likely cause or indicating the direction a solution should take, which the answerer then develops into a full answer. Either way, the answerer does not begin from a blank page, because they can borrow what is sound, follow the direction the AI answer establishes, and build on it.

This is where the law of least effort becomes central. People invest only the effort a goal requires and prefer the least costly route to reaching it (Kahneman 2011). A high-quality AI answer lowers the cost of producing a high-quality human answer, because part of the difficult work is already done, whether by providing adaptable material or by charting a viable direction. The easiest path to a strong contribution is therefore to take a strong AI answer as a foundation and then refine, extend, or follow through on it, whereas a weak AI answer provides no such shortcut. The higher the quality of the AI answer, the more an answerer can leverage, and the better the human answer they can produce at a given level of effort.

An answerer working from a sound AI answer thus inherits either a correct diagnosis and working code to verify and adapt or a well-aimed direction to develop, and in both cases can arrive at a demonstrably correct answer more readily than one working from a weak AI answer. This logic is consistent with competition research showing that a response is shaped by how much the competitor appears to have invested (Chen 2009; Hsieh et al. 2015), but the operative mechanism here is reuse and redirection: a better AI answer is a better foundation to build on.

H6. The effect of an AI answer on human answer quality becomes more positive as the quality of the AI answer increases.

4. RESEARCH CONTEXT

4.1. SegmentFault and the Randomized Field Experiment

SegmentFault is a prominent Chinese technical Q&A community specializing in IT and software development. Founded in 2012, it serves nearly 7 million users across web/mobile development, AI, cloud computing, and cybersecurity. As of April 2024, the platform hosted over 310,000 technical questions with an average of 1.9 answers each, and among global technical Q&A communities it ranks sixth in questions asked, third in answers provided, and first in answer-to-question ratio.[1] Its ecosystem relies on user-generated content: members pose and answer questions, and a reputation system awards points and medals based on the quantity and quality of their contributions.

We leverage a randomized field experiment conducted by the platform. Beginning September 13, 2023, SegmentFault ran an A/B test of its internal GenAI in which each newly posted question was independently and randomly assigned, upon posting, either to a treatment group that received an AI-generated answer or to a control group that received none, with no systematic variation by questioner characteristics, timing, or other factors.[2] This question-level randomization lets us isolate how internal GenAI affects human contribution while external GenAI remains available to all users. Online Appendix A provides further detail.[3]

Figure 3 illustrates a treated question and a control question. Once the system assigns a new posted question to the treatment group, it sends the question to an LLM to generate an answer.[4] The LLM computation may introduce a small delay, within minutes, between the question posting and the arrival of the AI answer. This delay is negligible relative to human response latency, as the first human answer appears after 22.067 hours on average. The AI answer is not interleaved with human answers according to posting time; instead, it is anchored to the top of the answer list. Community members may answer regardless of whether an AI answer is present. When one exists, the question page initially shows only its first two lines; clicking “View More” opens a pop-up window labeled “AI Answers” that carries a disclaimer warning of possible

[1] This information can be found at: https://stackexchange.com/sites?view=list#technology-questions.

[2] We confirmed with the SegmentFault’s operational manager who oversaw and the engineer who implemented the randomized experiment that the random assignment did occur at the question level.

[3] The Online Appendix is available at: https://researchbox.org/5255&PEER_REVIEW_passcode=UFACJZ.

[4] This information is provided by the SegmentFault’s operational manager overseeing this experiment.

errors and noting that the content is for reference only.

Two features of this deployment are central to our design. First, the AI answer is explicitly labeled, so the source signal in our framework is observable both to community members and to us. Second, it arrives before any human contributor, so it is the competitor a human answerer confronts on arrival rather than a supplement to an established discussion.

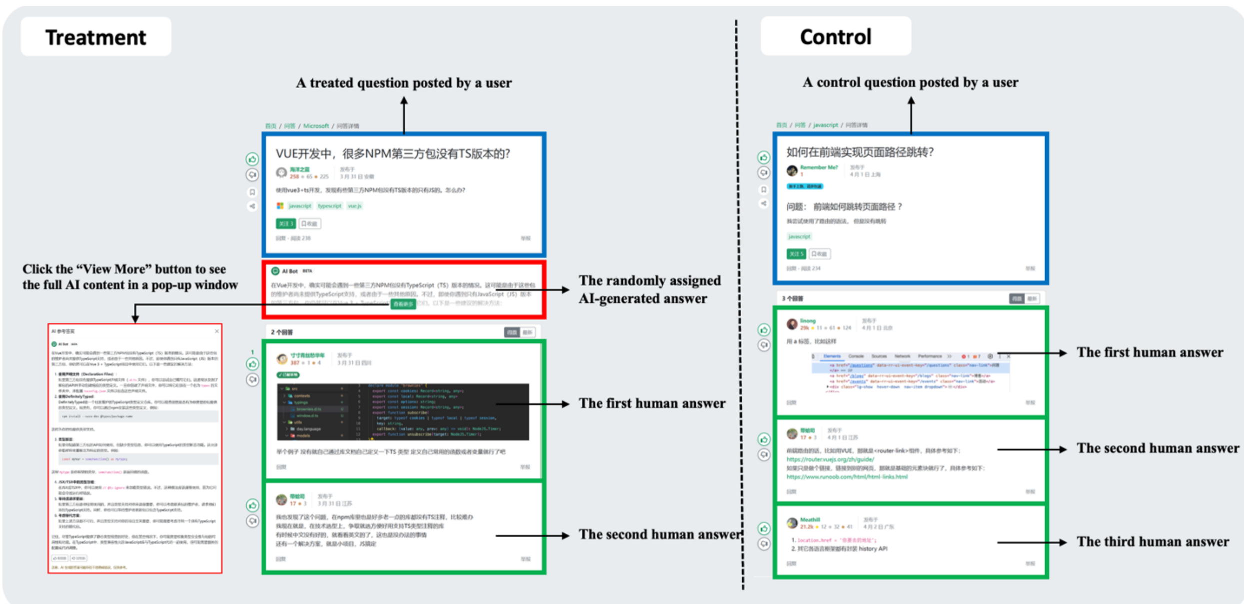


Figure 3. Illustration of treatment and control questions

4.2. Data Collection

We crawled data from SegmentFault in February 2024. Our main analyses focus on the 150-day window from September 2023 (when the field experiment began) to February 2024. We collected all questions posted during this timeframe. Our dataset comprises 3,613 questions posted by 1,404 unique users during this 150-day window, averaging 24 new questions daily. AI-generated answers appeared in 1,766 questions (48.9% of total). We collect additional pre-treatment data where needed for additional analyses (e.g., the spillover test in Section 6.4.3).

The platform assigns each question a tag as a content summary. Based on 384 unique tags, we classify questions into ten categories: web and mobile development, software design and architecture, cloud computing, databases, computer networks and systems, AI and machine learning, development tools and environment, programming languages, career and professional development, and other. "Web and mobile development" is the largest category, accounting for 57.5% of questions.

4.3. Summary Statistics

Table 2 reports summary statistics. Questions average 86.809 Chinese characters and receive 1.293 human answers, with 76.1% attracting at least one answer and 32.2% having an answer "accepted." Consistent with a negative average treatment effect on answer supply, treated questions receive fewer human answers than control questions (1.209 vs. 1.372). Questioner reputation scores range from −3 to 28,800, averaging 378.337.[5]

Table 2. Descriptive statistics of main variables

Variable	Mean and SD			Min	Max
	All samples	Treatment group	Control group		
question length (number of Chinese characters)	86.809 (113.981)	84.233 (89.347)	89.272 (133.323)	0.000	2702.000
number of human answers received by a question (full observation window)	1.293 (1.166)	1.209 (1.133)	1.372 (1.192)	0.000	8.000
percentage of questions that received at least one answer	0.761 (0.427)	0.724 (0.447)	0.795 (0.404)	0.000	1.000
percentage of questions that have an answer "accepted" by the questioner	0.322 (0.467)	0.293 (0.455)	0.350 (0.477)	0.000	1.000
reputation scores of the questioner at the time the question was posted	378.337 (1135.056)	334.484 (695.343)	420.267 (1433.522)	-3.000	28800.000

Notes: 1. Columns 2-4 present the means, with the standard deviations shown in parentheses.
2. The minimum question length of 0 represents questions where users posted content solely through images without accompanying text. These comprise 2.71% of our sample (98 out of 3,613 observations). Our main results remain unchanged when these observations are excluded.

4.4. Verifying Random Assignment

Our identification depends on the random assignment of questions to treatment and control. We validate it using t-tests and standardized differences (Austin 2009) across pre-assignment

[5] In SegmentFault, receiving downvotes on an answer can result in deductions to the answerer's reputation score, potentially pushing it into negative territory.

covariates: questioner characteristics (reputation; gold, silver, and bronze medal counts; total questions and answers; account age), questioner activity in the 1, 2, and 3 months before the experiment (accepted answers, answers provided, questions asked, and comments posted), and question content (character count and the presence of photos, hyperlinks, tables, and block quotes). We also use an LLM to score each question's linguistic complexity, technical-jargon use, and difficulty (Li et al. 2024; see Online Appendix C for LLM-based quality measure details and validation tests), and we account for question category and posting time.

The balance checks confirm successful randomization. All t-tests are statistically insignificant ($p > 0.05$), and all standardized differences fall well below the 0.10 threshold, with absolute values ranging from 0.003 to 0.061 (Online Appendix Table B1). Regression-based checks, which provide both an omnibus test of overall balance and covariate-level diagnostics while accounting for category and timing, reach the same conclusion (Online Appendix Table B2).

5. EMPIRICAL ANALYSIS

5.1. Main Variables of Interest

Human answer supply. Our first outcome is the volume of answers a question attracts, which reflects answerers' entry decisions. We measure it as the logarithm of the number of answers a question receives in seven days of being posted, denoted *Log(human answer number in 7 days)*.

Human answer quality. Our second outcome is the quality of human contributions. We employ an LLM to generate the quality measure. The detailed generation procedure and its validity assessment are provided in Online Appendix C. When studying human contribution quality, we apply this measure to the first human answer, denoted *First human answer quality*.

AI answer's profile. For treated questions, we characterize the AI answer along the two dimensions. *Length* is the logged character count of the AI answer, and it stands for the coverage signal of Section 3.2. *Quality* is the LLM-generated quality described above, applied to the AI answer, and it stands for the benchmark signal introduced in Section 3.2. Both are defined only where an AI answer exists and are set to zero for control questions.

5.2. Econometric Models

We model the impact of the AI answer as follows:

$$Y_i = \delta_0 + \delta_1 AI_i + X_i{}'\gamma + \delta_t + \varepsilon_i \quad (1)$$

where i indexes questions, Y_i is one of our two outcomes: the logarithm of the number of human answers received by question i in seven days, or the quality of the first human answer to question i, AI_i indicates treatment assignment (1 if AI-generated answer provided, 0 otherwise), X_i includes control variables (questioner characteristics, question content, and question type), and δ_t captures question posting date fixed effects. The impact of the AI answer, δ_1, is a function of its length and quality:

$$\delta_1 = \beta_1 + \beta_2 * Length + \beta_3 * Quality \quad (2)$$

where β_1 captures the level effect of the AI label, β_2 captures how the AI effect varies with its length, and β_3 captures how the AI effect varies with its quality. Substituting Equation 2 into Equation 1 yields:

$$Y_i = \delta_0 + \beta_1 AI_i + \beta_2(AI_i \times Length_i) + \beta_3(AI_i \times Quality_i) + X_i{}'\gamma + \delta_t + \varepsilon_i \quad (3)$$

In this specification, β_1 is what an AI answer does when both content dimensions are at zero, that is, the effect of an announced AI presence stripped of what the AI said. It provides the test of H1 (for answer supply) and H4 (for answer quality). The coefficients β_2 and β_3 show how the consequence of having an AI answer under a question changes as that answer's content changes. A negative β_2 means that a longer AI answer depresses human contribution more than a shorter one; a β_3 indistinguishable from zero means the consequence does not shift as the AI answer gets better. These interaction coefficients provide the tests of H2 and H5 (length) and of H3 and H6 (quality).

6. RESULTS

6.1. How the Effect on Answer Supply Varies with the AI Answer

Table 3 presents our main findings on the impact of internal GenAI-generated answers. Column 1 shows that the average effect of adding an internal GenAI answer on answer supply is negative and significant (−0.046, $p < 0.001$). We then examine in Column 2 how this effect varies with AI answer length and AI answer quality.

Table 3. The impact of AI-generated answers on answer supply and answer quality

	(1)	(2)	(3)	(4)
DV	Log(human answer number in 7 days)		First human answer quality	
AI_i	-0.046***	0.188*	0.072	0.411
	(0.011)	(0.080)	(0.083)	(0.718)
AI_i * $Length_i$		-0.046***		-0.204
		(0.014)		(0.128)
AI_i * $Quality_i$		0.004		0.114**
		(0.004)		(0.037)
question controls	√	√	√	√
questioner controls	√	√	√	√
question timing FE	√	√	√	√
N	3,613	3,612	2,727	2,726

Notes. 1. Standard errors in parentheses. . p<0.1, * p<0.05, ** p<0.01, *** p<0.001.
2. Column 2 has fewer observations than Column 1 because the LLM declined to provide the assessment for one AI answer. Columns 3 and 4 are estimated on questions with a first human answer and the required first-human-answer-quality outcome. Column 4 additionally requires usable AI-answer quality and therefore excludes one further observation.

The label effect (H1). The AI indicator is positive and significant (0.188, $p < 0.05$). Projected to an AI answer whose content contributes nothing, the presence of a labeled AI answer raises rather than lowers the number of human answers a question attracts, consistent with contributors treating the AI as a weak competitor whose appearance invites a response. This provides support for H1.

The length effect (H2). The interaction between the AI indicator and AI answer length is negative and significant (−0.046, $p < 0.001$). As the AI answer gets longer, the consequence of having it under the question becomes more adverse to human contribution. This provides support for H2.

The quality effect (H3). The interaction with AI answer quality is small and statistically indistinguishable from zero (0.004, $p = 0.391$). The consequence of having an AI answer under a question does not depend on how good that answer is. This is the prediction that separates our account from a capability-deterrence account, under which better AI should deter more. We find no evidence against the null prediction of H3.

6.2. How the Effect on Answer Quality Varies with the AI Answer

Columns 3 and 4 of Table 3 examine the effect of internal GenAI-generated answers on the quality of the first human answer. Column 3 shows that the average effect of internal GenAI on the quality of the first human answer is small and statistically insignificant (0.072, $p = 0.381$), and we proceed to examine how this average effect is composed through the AI content profile in Column 4.

The label effect (H4). The AI indicator is not statistically significant (0.411, $p = 0.495$). Consistent with H4, the bare fact of machine authorship does not shift the quality of the human answer. Once a contributor has committed to answering and engages with the content, their answer is calibrated to the quality of the AI answer rather than to the abstract source cue. As with our other null predictions, we treat this as an absence of evidence for a label effect on quality rather than proof of an exact zero.

The length effect (H5). The interaction with AI answer length is negative but not statistically significant (−0.204, $p = 0.054$). How much of the question the AI answer seems to cover does not change how good the human answer is, which is consistent with our theoretical prediction. We find no evidence against H5.

The quality effect (H6). The interaction between the AI indicator and AI answer quality is positive and significant (0.114, $p < 0.01$): the higher the quality of the AI answer, the higher the quality of the human answer that follows. This is consistent with our account, in which a stronger AI answer gives the contributor a better foundation to build on and raises the quality of the resulting human answer. H6 is therefore supported.

6.3. The Pattern as a Whole

Table 4 sets the six tests against their predictions, and the estimates trace the pattern predicted by our framework in Section 3. Length reduces supply (H2: −0.046, $p < 0.001$), and quality raises effort (H6: 0.114, $p < 0.01$). The two predictions of null effect behave as our account requires: we find no evidence that quality shifts supply (H3) or that length shifts effort (H5). We consider these as an absence of evidence for an effect, not as proof that the effect is exactly zero. The label raises answer supply (H1: 0.188, $p < 0.05$) and, consistent with H4, has no detectable effect on human answer quality. Read together, the cues an answerer registers at a glance, including the label and the length, govern entry; while the cue that requires reading, quality, governs the crafting decision.

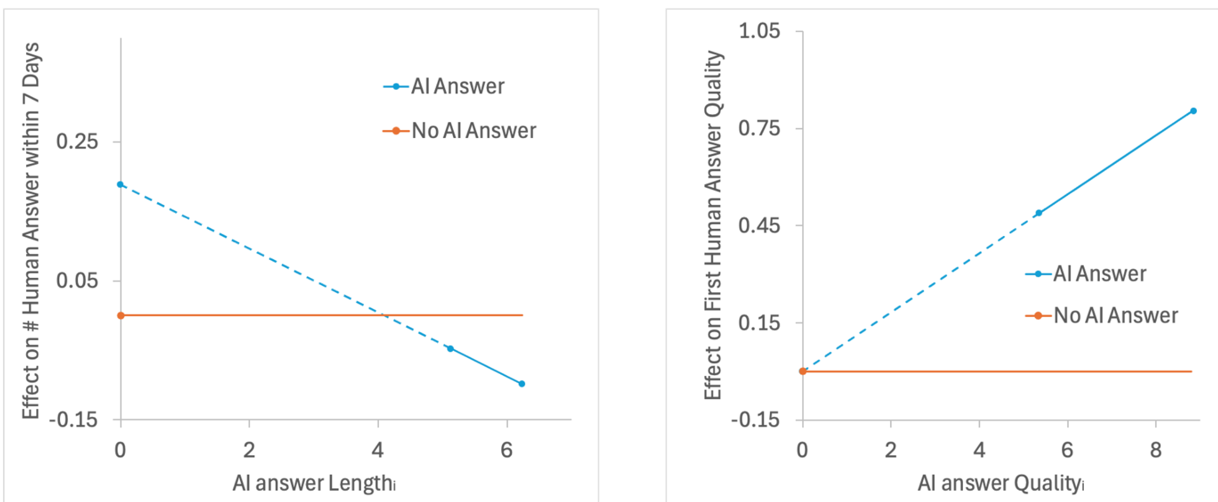
Table 4. Predictions and results

	Answer supply (entry)	**Answer quality (effort)**
Label	H1 (positive): 0.188*, Supported	H4 (no effect predicted): n.s., consistent
Length	H2 (negative): −0.046***, Supported	H5 (no effect predicted): n.s., consistent
Quality	H3: (no effect predicted): n.s., consistent	H6 (positive): 0.114**, supported

Notes. · p<0.1, * p<0.05, ** p<0.01, *** p<0.001.

The results show a clear separation between System 1 and System 2 processing in the entry and crafting decisions. The insignificant result for H3 rules out the possibility that higher-quality AI answers lower human supply. What remains is that quality can be judged only by reading, and reading follows the decision to enter, so the quality of the AI answer shapes the human answer (H6).

Figure 4 visualizes these two gradients. Panel (a) plots the effect on answer supply as AI answer length varies, and Panel (b) plots the effect on answer quality as AI answer quality varies. In each panel, the horizontal line at zero is the no-AI-answer baseline, the solid segment covers the range within one standard deviation of the mean of the varying dimension, and the dashed segment extends toward zero content to show the projected label effect. Panel (a) shows a line that falls below the no-AI-answer baseline over most of the observed range, driven mainly by answer length; Panel (b) shows a line that lies above the baseline, rising with AI answer quality.



(a) Answer supply:
label motivates, length deters

(b) Answer quality:
higher AI quality raises human quality

Figure 4. The effect of AI answers depends on the AI answer's profile.

These results also show a platform where its choices matter. Over the range we observe, it is length, not quality, that suppresses entry. The practical implication is that the benefits of quality are realized when the AI system is made better without being made longer.

This distinction is important because length and quality often rise together for the wrong reason. AI-generated text can be padded rather than substantive: hedging preambles, restated questions, and formulaic summaries add length without adding correct or novel content, a phenomenon increasingly described as “AI slop.” Our two gradients suggest that such padding is the worst of both worlds for a knowledge community. It does not raise the quality that improves human answers (H6), but it adds the length that displaces entry (H2/H3), because the question now looks answered. Verbose but empty AI answers thus deter participation while delivering none of the offsetting benefit of quality.

The design implication is therefore sharper than “improve the AI.” A platform should raise the informational density of the AI answer, its accuracy and coverage per unit of length, rather than its volume, and should be wary of systems whose apparent thoroughness is mostly formulaic bulk. Concise, high-quality answers capture the gains of quality while minimizing displacement.

6.4. Robustness Checks

6.4.1. Alternative Specifications

Our main analysis uses log-transformed count data for the number of answers per question. Recent research has raised concerns about log-transforming non-negative dependent variables (Chen & Roth 2024). We therefore re-estimate the two main relationships under models that do not require this transformation.

Answer Supply. Table 5 examines the human answer supply across four additional specifications. Column 1 adds a questioner random effect. Because treatment is randomized at the question level, the error term is uncorrelated with treatment, so the random effect is an appropriate adjustment for repeated questions from the same questioner. Columns 2-4 employ models specifically designed for count data: Poisson, zero-inflated Poisson (ZIP), and zero-inflated negative binomial (ZINB).

The three coefficients of interest are stable across all specifications. The AI label effect is positive and significant throughout. It ranges from 0.168 ($p < 0.05$) under the random effect model to 0.469 ($p < 0.05$) under the zero-inflated negative binomial model, and the three count models sit close to 0.468. The interaction with AI answer length is negative and significant in every column, from -0.041 ($p < 0.01$) under the random effect model to -0.117 ($p < 0.01$ or better) under the count models. The interaction with AI answer quality is small and never significant, moving only between 0.003 and 0.010. Adding the questioner random effect in Column 1 leaves the OLS estimates almost unchanged. This pattern matches H1, H2, and H3.

Table 5. Robustness checks: alternative model specifications for human answer supply

	Random Effect	Poisson	ZIP	ZINB	Random Effect
	(1)	(2)	(3)	(4)	(5)
	DV: Log(human answer number in 7 days)	DV: Human answer number in 7 days			DV: First human answer quality
AI_i	0.168*	0.468*	0.468*	0.469*	0.410
	(0.081)	(0.199)	(0.224)	(0.224)	(0.603)
*AI_i * $Length_i$*	-0.041**	-0.117***	-0.117**	-0.117**	-0.204·
	(0.014)	(0.034)	(0.039)	(0.039)	(0.106)
*AI_i * $Quality_i$*	0.003	0.010	0.010	0.010	0.114**
	(0.004)	(0.014)	(0.014)	(0.014)	(0.036)
other controls	√	√	√	√	√
question timing FE	√	√	√	√	√
questioner random effect	√				√
N	3,612	3,612	3,612	3,612	2,726
AIC	2741.982		10108.205	10110.206	11959.154
BIC	3924.657				13088.076
log likelihood	-1179.991	-4864.102	-4864.102	-4864.103	-5788.577
deviance		3537.101			
pseudo R-squared		0.014			

Notes. 1. Standard errors in parentheses. . $p<0.1$, * $p<0.05$, ** $p<0.01$, *** $p<0.001$.
2. In Column 1 the dependent variable is the log of the number of human answers provided within 7 days. In Columns 2-4 the dependent variable is the raw count of human answers provided within 7 days. In Column 5 it is the quality of the first human answer.

Human Answer Quality. Column 5 in Table 5 reports the OLS estimate and a specification with a questioner random effect. The interaction between the AI indicator and AI answer quality stays positive and significant in both columns (0.114, $p < 0.01$). The questioner random effect does not alter the conclusion.

6.4.2. Alternative Time Frames

Table 6 examines robustness to alternative time frames. Columns 1-3 vary the window for counting human answers from 7 days in the main model to 3 days, 14 days, and the full observation period after question posting. The results are consistent across windows. The AI label effect is positive and significant, and the interaction with answer length is negative and significant in all columns. The interaction with answer quality is not significant throughout.

Columns 4-6 restrict the sample to questions posted within the first 30, 60, and 90 days after the experiment began. Statistical power considerations are critical for interpreting these results. Our main dependent variable (number of human answers within 7 days) has a mean of 1.136 (SD: 1.093) in the treatment group and 1.305 (SD: 1.172) in the control group. Detecting the observed length slope of -0.040 with 80% power and $\alpha = 0.05$ requires approximately 14,764 questions for a simple t-test. Even our regression specification includes a rich set of controls that absorb a substantial portion of residual outcome variance, which increases the precision of coefficient estimates, samples that are too small remain insufficient for reliable inference.

The 30-day window (Column 4, N=837) produces an insignificant coefficient (-0.041, $p < 0.10$), which is likely due to insufficient statistical power. The 60-day and 90-day windows (Columns 5-6, N=1,606 and 2,412) provide adequate samples and reveal significant negative effects (-0.050 and -0.044, $p < 0.01$).

Table 6. Robustness checks: alternative time frame used in analyses

	(1)	(2)	(3)	(4)	(5)	(6)
	DV: Log(human answer number in N days)			DV: Log(human answer number in 7 days)		
	N=3	N=14	N=All	30 days sample	60 days sample	90 days sample
AI_i	0.167*	0.185*	0.236*	0.190	0.235*	0.212*
	(0.080)	(0.078)	(0.115)	(0.127)	(0.105)	(0.090)
$AI_i * Length_i$	-0.040**	-0.047***	-0.067***	-0.041·	-0.050**	-0.044**
	(0.014)	(0.014)	(0.020)	(0.023)	(0.019)	(0.016)

AI_i * $Quality_i$	0.002	0.004	0.011	0.002	0.001	-0.000
	(0.004)	(0.004)	(0.007)	(0.009)	(0.007)	(0.006)
question timing FE	√	√	√	√	√	√
other controls	√	√	√	√	√	√
N	3,612	3,612	3,612	837	1,606	2,412

Notes. 1. Standard errors in parentheses. . p<0.1, * p<0.05, ** p<0.01, *** p<0.001.
2. In Columns 1-3 the dependent variable is the log of the number of human answers provided within the indicated window. In Columns 4-6 the dependent variable is the log of the number of human answers provided within 7 days, and the sample is restricted to questions posted within the first 30, 60, and 90 days after the experiment began.

Table 7 applies the same subsamples to the quality outcome. The interaction between the AI indicator and AI answer quality is positive in all three subsamples (0.110, 0.106, and 0.149), and its significance rises with the sample size. It is insignificant in the 30-day sample (N = 592), marginally significant in the 60-day sample (0.106, $p < 0.1$), and significant in the 90-day sample (0.149, $p < 0.01$). The positive sign is stable across all three, and the increase in significance with sample size follows the same power logic that governs the supply results. The choice of observation window does not overturn the benchmark relationship in H6.

Table 7. Robustness checks: alternative time frame for first human answer quality

	(1)	(2)	(3)
DV: First human answer quality	30 days sample	60 days sample	90 days sample
AI_i	-0.216	0.346	0.320
	(1.028)	(0.861)	(0.700)
AI_i * $Length_i$	-0.059	-0.180	-0.230·
	(0.194)	(0.155)	(0.124)
AI_i * $Quality_i$	0.110	0.106·	0.149**
	(0.074)	(0.057)	(0.045)
other controls	√	√	√
question timing FE	√	√	√
N	592	1,214	1,833

Notes. 1. Standard errors in parentheses. . p<0.1, * p<0.05, ** p<0.01, *** p<0.001.
2. The sample is restricted to questions observed for at least 30, 60, and 90 days.

6.4.3. Potential Spillover Effects

On SegmentFault, AI-generated answers are visible to all users and could potentially divert attention from questions without AI answers. This contamination might influence outcomes for control questions based on the treatment status of other questions. Such spillover effects would violate the stable unit treatment value assumption (SUTVA), which requires that potential outcomes depend exclusively on a unit's own treatment status, a fundamental condition for valid causal inference in randomized experiments.

To test for potential spillover effects, we implement a comparative analysis between the five-month data of our control group and historical data from five months before the introduction of AI-generated answers (May 2023-September 2023). The two groups are both questions without AI answers and are directly comparable. Pre-experiment questions have been on the platform longer and have mechanically accumulated more answers over time. Our spillover test addresses this concern through regression controls for a cubic polynomial of question age (to substitute question/answer timing fixed effects in Equation 1, which is not applicable to the two groups on two different time slots). If spillover effects exist, control questions during the experiment period should show different outcomes compared to similar questions from the pre-AI period, as the control questions would be indirectly affected by AI answers to treated questions. Table 8 shows no significant differences between our control group and pre-AI data for answer supply, suggesting spillover is not a concern.

Table 8. Robustness to potential spillover effects

	(1) DV: Log(human answer number in 7 days)
post-AI deployment control group indicator	-0.022 (0.015)
controlling for a cubic polynomial of question/answer age	√
other controls	√
N	5,984

Notes: 1. Standard errors in parentheses. . p<0.1, * p<0.05, ** p<0.01, *** p<0.001.
2. We also applied all robustness checks conducted for the main effects to the spillover tests. All results remain consistent across specifications, showing no significant spillover effects.

6.4.4. Quality of Random Assignment

The platform assigns AI answers to questions at random. To further strengthen confidence in the randomization, we assess whether our estimates could be sensitive to selection on unobservable characteristics, using the method of Oster (2019).

This approach examines how the treatment effect changes with observable controls and infers potential changes if unobservables were controlled. The method provides two key parameters: δ, which indicates how important unobservables would need to be relative to observables to nullify the treatment effect (values exceeding 1 suggest robustness), and β^*, the bias-adjusted effect assuming selection on unobservables. These parameters require an assumption on R^2_{max}, the maximum possible R-squared value achievable if all unobservables were controlled for. R^2_{max} is typically set at 1.3 times the R^2 from the regression with observable controls only (Oster 2019).

Table 9 reveals remarkable consistency between original and bias-adjusted coefficients. For answer supply, the original coefficient (-0.046) differs negligibly from the bias-adjusted coefficient (-0.045). The δ values for the outcome (39.719) far exceed the recommended threshold of 1, indicating that unobservables would need to be dramatically more important than observables to explain away our findings. These results provide strong evidence for the quality of random assignment and robustness of our main conclusions.

Table 9. Robustness to the quality of random assignment

	Original coefficient $\hat{\beta}$	R^2	$R^2_{\max}$	Bias-adjusted effect β^*	δ for β=0 given $R^2_{\max}$
DV: Log(human answer number in 7 days)	-0.046	0.093	0.121	-0.045	39.719

Note: $\hat{\beta}$ and R^2 refer to our OLS coefficients and corresponding R-squared values. β^* is the bias-adjusted treatment effect. δ indicates how important unobservables would need to be, relative to observables, to explain away the estimated treatment effect.

6.4.5. Bounding the Effect of AI Answer Quality Under Sample Selection

To estimate the effect of AI answer quality, we restrict the analysis to the first human answer (Table 3 Columns 3 and 4). This analysis conditions on a post-treatment outcome, because a question enters the sample only if it received at least one human answer. If the questions that attract a human answer differ systematically from those that do not, the estimate of AI answer quality's effect (β_3) can be biased. To check whether our conclusion survives this concern, we compute Lee (2009) bounds, which give a range that is guaranteed to contain the true effect even under this kind of selection.

The Lee method is built for a binary treatment, but AI answer quality is a continuous measure. We therefore convert the continuous measure into a binary comparison in two different ways, so that the conclusion does not depend on any single choice.

The first approach compares high- versus low-quality answers within the treated sample. We split treated questions at the median of AI quality into a high group and a low group, apply the Lee bounds to this two-group comparison, and then divide the resulting range by the gap in average quality between the two groups. This converts the bound on the group difference into a bound on the per-unit slope β_3.

The second approach uses finer quality bins and compares them to the control group. We sort treated questions into terciles of AI quality, compute Lee bounds for the top and bottom terciles relative to the control group, and take the difference between these two bounds divided by their gap in average quality. Because it uses the extremes of the quality distribution rather than a single median cut, this approach draws on more of the continuous variation in quality.

Both approaches produce ranges that lie entirely above zero ([0.047, 0.402] for the first approach and [0.042, 0.337] for the second). This means that even in the most pessimistic scenario allowed by the selection problem, higher AI answer quality is still associated with higher human answer quality. Because the entire range excludes zero, the positive effect cannot be explained away as an artifact of which questions happen to receive a human answer, and the two independent bounding strategies further reinforce this conclusion. Appendix D provides additional details for the estimation.

6.5. Mechanisms

In this section, we probe the mechanisms that underlie our hypotheses. The goal is not to measure the mechanisms directly, since they are not observed, but to test their observable implications. Each mechanism implies patterns that should appear in the data if it is operating, and often patterns that differ from those implied by alternative explanations. For each hypothesis, we test for these patterns and use them, where possible, to distinguish our account from alternative explanations. These results should be read as support for the plausibility of our arguments, not as estimates of the magnitude of each channel.

We focus on the hypotheses that predict a directional effect and set aside those predicting a null. A directional prediction rests on a specific channel, and that channel implies further, testable patterns we can probe or use to rule out alternatives. We therefore examine the three supported hypotheses, H1, H2, and H6, in turn: for H1 we probe the weak-competitor account, for H2 we test why length reduces contribution, and for H6 we test why higher AI quality raises human contribution quality.

6.5.1. Is the AI Appraised as a Weak Competitor? (H1)

Our account for the positive label effect is that users perceive the AI as a weak competitor and are therefore willing to step in and answer. We present a test on the AI as weak competitor account. The key idea behind this test is that the weak-competitor perception is not equally relevant for every question. It becomes salient precisely when a question is hard. Easy questions can be answered well by almost anyone, a GenAI system included, so whether the first answer comes from a strong or a weak source makes little difference: a competent answer is a competent answer, and there is little for a human to add. Hard questions are different. Answering them well requires the kind of specialized expertise, firsthand experience, or situation-specific context that a general AI is widely believed to lack. For example, questions that turn on professional judgment in an unusual edge case, on hands-on experience with a specific tool or procedure, or on niche, rapidly changing knowledge that a generic model is unlikely to have. On exactly these questions, two things happen at once: the belief that the AI is only a weak competitor is more likely to be true, and it is more likely to be *top of mind*.

This yields a sharp, testable implication. If the label effect operates through the weak-competitor belief, that effect should concentrate on difficult questions. We measure question difficulty using an LLM–generated score, with details in the Appendix C, and we split the sample at the mean difficulty. Table 10 confirms this prediction. On difficult questions the AI label effect is positive and significant (0.282, $p<0.01$), and the interaction with answer length is negative and significant (-0.059, $p<0.001$). On easy questions the AI label effect is not significant.

Table 10. The effect on answer supply by question difficulty (for H1)

	(1)	(2)
DV: Log(human answer number in 7 days)	difficult questions	easy questions
AI_i	**0.282****	-0.046
	(0.099)	(0.134)
$AI_i * Length_i$	**-0.059*****	-0.017
	(0.017)	(0.024)
$AI_i * Quality_i$	0.002	0.012
	(0.005)	(0.008)
question controls	√	√
questioner controls	√	√
question timing FE	√	√
N	2,377	1,235

Notes. 1. Standard errors in parentheses. . $p<0.1$, * $p<0.05$, ** $p<0.01$, *** $p<0.001$.
2. Questions are split at the mean of the difficulty score. Splitting at the median produces similar results.

6.5.2. Why Length Reduces Contribution (H2)

The length slope shows that as the AI answer gets longer, its presence under the question becomes more of a deterrent to human contribution. In our theoretical development we argued that human answerers treat length as a proxy for coverage (i.e., how much ground the AI answer addresses), and stay out when little untouched ground appears to be left. This section probes that account against the most natural alternative.

The alternative is that length proxies for quality rather than coverage. A longer answer seems more thorough and authoritative, which signals a high bar that would be hard to outperform. Both accounts are consistent with the competition lens and both work through length.

The fast nature of the entry decision favors coverage. Length is plausibly informative about coverage. A longer, more heavily itemized answer looks like it touches more distinct points. However, it is uninformative about quality, which can be judged only through the careful reading that a fast entry decision does not involve. An answerer glancing at length can form an impression of how much ground is covered, but not of whether that ground is covered correctly. This yields separable predictions. If length proxies for coverage, then actual coverage should carry the deterrent, and accounting for it should absorb the length effect; if length proxies for quality, then actual quality should do so.

Our main results already speak against the proxy for quality explanation. The actual quality of the AI answer is not associated with the entry decision; if answerers were using length to proxy for quality, higher actual quality would be linked to less entry, and it is not. We then test the proxy for coverage account directly, measuring the AI answer coverage as the count of distinct points the answer lays out.

Table 11 sets out the logic. Columns 1 and 2 reproduce our main results. Column 1 includes only the interaction with length, which is negative and significant (−0.044, $p<0.01$). Column 2 adds the interaction with quality. If length were a proxy for quality, accounting for quality should absorb the length effect; instead the length interaction is essentially unchanged (−0.046, $p<0.001$) and the quality interaction is insignificant, so length is not standing in for quality. Column 3 adds the interaction with coverage. Here the length interaction falls in magnitude and to nonsignificance (−0.028, p<0.1) and coverage is negative and significant (−0.007, p<0.05). This is what it looks like for a proxy to be resolved: once the quantity that length stands in for is measured, the deterrent attaches to that quantity, and length's own coefficient absorbs into it. Length deters because answerers use it as a proxy for coverage.

One might ask whether length and coverage are simply the same quantity measured twice, which would make the test vacuous. They are not: their correlation is only moderate (correlation = 0.448), because AI answers can be padded with filler or repetition that adds length without adding ground. This has two implications. First, it is what makes the test informative; if length and coverage moved in lockstep, no test could reveal which quantity the answerer is proxying for. Second, the asymmetry is why length proxies for coverage rather than quality: at a glance, length remains a positive, if imperfect, indicator of how much ground is covered, whereas it says nothing about whether that ground is covered correctly. The answerer glances at how far the answer runs, and given the structured format of these AI answers, that glance stands in for roughly how much has already been said.

Table 11. Why length reduces answer supply (for H2)

	(1)	(2)	(3)
DV		Log(human answer number in 7 days)	
AI_i	0.202**	0.188*	0.129
	(0.078)	(0.080)	(0.087)
AI_i * $Length_i$	-0.044**	-0.046***	-0.028.
	(0.014)	(0.014)	(0.016)
AI_i * $Quality_i$		0.004	
		(0.004)	
AI_i * $Coverage_i$			**-0.007***
			(0.003)
question controls	√	√	√
questioner controls	√	√	√
question timing FE	√	√	√
N	3,613	3,612	3,613

Notes. 1. Standard errors in parentheses. . p<0.1, * p<0.05, ** p<0.01, *** p<0.001.
2. Column 2 has fewer observations because the LLM declined to provide the assessment for a specific AI answer.

6.5.3. Why Higher AI Quality Raises Contribution Quality (H6)

The main results show that a higher-quality AI answer is followed by a higher-quality human answer. Our account is reuse and redirection: a stronger AI answer lowers the cost of producing a strong human answer because it hands the contributor something to build on (e.g., sound material to adapt or a viable direction to develop), so the answer a given contributor can produce at a given level of effort improves as the AI answer improves.

Reuse and redirection is not the only way to read this result. Two alternatives could produce the same aggregate pattern. The first is *selection*: a higher-quality AI answer may draw contributions from more expert users, so the observed rise reflects who shows up rather than how any given person writes. The second is *differentiation*: rather than building on the AI answer, humans may steer away from it, competing by providing what the AI does not, such as personal experience or a novel angle. The three accounts make different, testable predictions, and we use three pieces of evidence to separate them: which dimension of the human answer rises with AI quality, whether more expert contributors appear, and whether human answers move toward or away from the AI.

Which dimension rises pins down the mechanism. Reuse and differentiation disagree about where in the human answer the gain should appear. Reuse predicts the gain should concentrate on accuracy, because what a contributor can actually borrow or build on from a strong AI answer is its correct technical content, including a sound diagnosis, working code, the right direction, and reusing that material is precisely what makes the resulting human answer more accurate. Differentiation predicts the opposite: if humans compete by distinguishing themselves from a strong AI answer, the gains should land on dimensions where humans hold a distinctive edge and the AI offers nothing to reuse, such as personal experience or alternative solutions. To separate the two, we use an LLM to score ten quality dimensions of the first human answer: clarity, readability, accuracy, relevance, detail level, and the presence of personal experience, insight, innovative approaches, alternative solutions, and engaging storytelling (see Online Appendix C for definitions and validation). We regress each dimension on AI quality. Only one dimension responds. As Column 1 of Table 12 shows, the interaction between AI_i and answer quality is positive and significant on accuracy (0.083, $p < 0.05$), while for the other nine dimensions AI quality has no significant effect. A higher-quality AI answer is followed by a more accurate human answer, and by nothing else. This is exactly the signature reuse predicts, and it weighs against differentiation.

Ruling out selection. If selection drove the result, a higher-quality AI answer should attract more expert first contributors, and the quality gain would follow from their expertise rather than from any change in behavior. We measure the expertise of the first human contributor by the number of previously accepted answers within the relevant tag and regress it on AI quality. Column 2 of Table 12 shows no such relationship: the interaction between AI_i and answer quality is insignificant (-0.005), as are the other coefficients. Higher AI quality does not pull in more expert contributors, so the accuracy gain in Column 1 reflects how a given contributor responds to the AI, not a change in who responds.

Table 12. Effect on first human answer expertise and accuracy (H6, selection test and evidence from quality dimension)

DV	(1) First human answer accuracy	(2) First human expertise
AI_i	0.852	0.048
	(0.726)	(0.519)
AI_i * $Length_i$	-0.239·	0.011
	(0.129)	(0.091)
AI_i * $Quality_i$	0.083*	-0.005
	(0.041)	(0.032)
question controls	√	√
questioner controls	√	√
question timing FE	√	√
N	2,728	2,747

Notes. 1. Standard errors in parentheses. . p<0.1, * p<0.05, ** p<0.01, *** p<0.001.
2. The sample includes questions with a first human answer.

Ruling out differentiation. The dimension evidence already points away from differentiation; a second test confirms it from a different angle by examining how similar the human answer is to the AI answer. Differentiation predicts that if humans compete by giving answers that differ from the AI, then a better AI answer should make the human answer less similar to it. We measure the semantic similarity between the AI answer and the first human answer using Sentence-BERT. Table 13 supports reuse and redirection. The coefficient on $AIAnswerQuality_i$ is positive and significant (0.011, $p < 0.001$): as the AI answer improves, the human answer grows closer to it. This is the direct opposite of what differentiation predicts.

Table 13. Effect on similarity between the AI and human answer (H6, test differentiation)

DV	(1) Similarity (AI Answer, Human Answer)
$AIAnswerLength_i$	0.005
	(0.011)
$AIAnswerQuality_i$	**0.011*****
	(0.003)
question controls	√
questioner controls	√
question timing FE	√
N	1,277

Notes. 1. Standard errors in parentheses. . p<0.1, * p<0.05, ** p<0.01, *** p<0.001.
2. Similarity is computed with Sentence-BERT. The sample includes questions with at least two answers.

7. DISCUSSION AND CONCLUSIONS

7.1. Main results

Using a randomized field experiment at a leading technical Q&A community, we examine how a labeled, platform-provided internal GenAI answer affects human contribution when external GenAI remains available. We separate the total effect of an AI answer into the effect of its label and the effects of its content, and we distinguish two decisions a human answerer makes: whether to answer and how good that answer will be.

For answer supply, the AI label raises human contribution. This is consistent with a skeptical community that treats the machine as a weak competitor whose presence invites participation. Longer AI content, by contrast, suppresses contribution, because it signals that much of the question has been covered. Because typical AI answers in our setting are substantial, the length effect dominates the label effect, and the average effect on supply is negative. The quality of the AI answer does not shift supply.

For contribution quality, neither the label nor the length of the AI answer matters. A higher-quality AI answer raises the quality of the first human answer, because it gives the contributor sound material to adapt and a viable direction to develop.

Our mechanism tests support this account. The label effect concentrates on difficult questions, where the perception of AI as a weak competitor is more salient and relevant. Coverage rather than perceived quality drives the length effect. Higher AI quality raises human answer quality through reuse rather than through the selection of stronger contributors or through differentiation from the AI answer. Two findings make this clear: the gain concentrates on accuracy, the dimension a contributor can borrow and the dimension that is fundamental on a technical platform, and human answers grow more similar to stronger AI answers rather than more distinct from them.

7.2. Theoretical contribution

We contribute to the literature on GenAI in online communities by investigating platform-provided internal GenAI as a governance arrangement distinct from external GenAI, and by examining its consequences for human contributors when external GenAI remains available. Prior work often bundles the effect of AI content with the effect of the AI label. Our setting makes both the source and the content of the AI contribution visible, so we can estimate the two effects separately and test the distinct decisions through which each operates. We show that a community's response to an AI competitor is not a single quantity but depends on identifiable features of the AI answer, and that these features act through the specific level of processing at which they become available.

We also extend research on how prior contributions shape subsequent contribution. This literature establishes how the presence, volume, quality, and source of existing content affect a newcomer's decision, but it was developed almost entirely on human prior contributors. These features do not carry the same meaning when the prior contributor is an identified machine. The source cue invites entry rather than foreclosing it, and the same length and quality that would be read one way from a trusted human are read differently when the community regards the author as a weak competitor. Separating the label from the content makes this point demonstrable, and it is not available to designs in which the AI is an invisible drafting aid or a private substitute. It also distinguishes our account from a substitution view. Pure substitution predicts only decline, whereas an AI competitor whose content can be reused opens the route by which a better AI answer leads humans to write better rather than less.

The decomposition also carries a forward-looking benefit. Because our results tie the total AI effect to the content characteristics of the AI answer rather than to a fixed average, they provide a basis for reasoning about settings we do not observe. As AI capabilities advance and the properties of AI answers shift, one can combine the AI label effect with the corresponding AI content effect to anticipate how AI will reshape human contribution under the same competition mechanism. The label's role remains contingent on community beliefs that may themselves change.

The most direct point of comparison is Su et al. (2025), who study Quora's internal GenAI and report positive effects on both the quantity and the length of human contribution, whereas we find that a longer AI answer deters human contribution and that the average effect on supply is negative. The two sets of results look contradictory on the surface, but they are two readings of the same mechanism under different conditions. The AI label effect dominates in Su et al.'s (2025) setting, and the deterrent role of length does not materialize there.

The reason lies in whether contributors read the length of the AI answer as a signal of coverage when they decide whether to enter. In our technical setting, contributors treat length as a fast, surface cue for how much of the question has already been addressed, and a question that appears substantially addressed offers less perceived room for a new contribution. Length suppresses entry. Quora, by contrast, is a general-interest platform whose content is largely experiential and open-ended, and where many distinct answers can coexist and be valued at once. There, the value of a human contribution rests on personal experience and perspective that additional AI length may not foreclose, so length may not be read as a coverage cue. The deterrent channel that dominates our setting is muted, and the positive label effect prevails.

Timing compounds this contrast. In Su et al. (2025), the AI answer is introduced long after the original question was posted (average pre-policy tenure is roughly 49 months), by which point at least one human answer is already present (Su et al. 2025, pp. 15, 22). So the AI enters as a retrospective supplement to an established discussion rather than as the competitor a contributor confronts on arrival. In our setting the AI answer is generated quickly after the question posting and before human contributors arrive, so it is the rival a contributor evaluates at the moment of entry. This configuration activates the fast entry decision and the slower crafting decision we theorize, whereas a retrospective supplement largely does not.

Taken as a whole, the comparison supports the broader claim of this paper. Whether an identified AI contributor complements or displaces human contribution is not a fixed property of internal GenAI but a joint function of the appraisal the AI invites and the signals its content carries.

Reading both settings through the AI label effect and the AI content effect converts an apparent conflict in the literature into a single conditional prediction, under which an internal AI may act as complement or competitor depending on the domain, on how much content the AI provides, on how readily that content is read as coverage, and on when the AI answer arrives relative to human contributors.

7.3. Practical implication

Platforms that cannot eliminate external GenAI can still design an internal GenAI system, and our findings speak to how that design should proceed. The central lesson is that the label and the content of an internal AI answer are separate design choices that move human contribution in different ways. A visible AI label is not inherently harmful and can draw people in, above all on difficult questions where a weak competitor leaves room to add value, so platforms do not need to conceal that an answer is machine-generated.

The deterrent force comes from content that reads as already covering the question. Platforms that want to sustain answer volume should therefore be cautious about placing long AI answers in the salient position, and might present that content in a shorter or summarized form so it does not appear, at a glance, to have filled the solution space. A higher-quality AI answer works in the opposite direction. It hands contributors sound material and a direction to follow, so improving the internal AI raises the quality of human answers.

The managerial upshot is that the two design choices should be tuned in opposite directions, that is, hold visible length down while pushing accuracy and useful coverage up. This combination captures the reuse gains that lift human quality while limiting the displacement that longer answers cause, which allows a platform to integrate a capable AI and preserve the distinctive value of human contribution.

7.4. Limitations

Several limitations qualify our conclusions and point to future work. First, our evidence comes from a single technical Q&A community, where contributions can be evaluated on correctness and where practitioners at the time held domain-specific doubts about AI capability. As our comparison with Su et al. (2025) makes clear, the sign of the aggregate effect depends on the domain and on the timing of the AI answer's entry, so the specific magnitudes we report may not generalize to other online platforms or to settings where AI enters after human answers have accumulated. Our framework is intended to travel across these settings, but its predictions in domains unlike ours remain to be tested directly.

Second, the appraisal of AI as a weak competitor is historically situated, and as beliefs about AI capability continue to evolve, the AI label effect may change; our decomposition anticipates this shift but cannot by itself pin down when it will occur.

Third, the question-level randomization allows us to identify effects on human answering behavior but precludes any statement about question supply. Future research could examine how these effects evolve as AI capability and community beliefs change, extend the label-versus-content decomposition to other domains and other forms of platform-integrated AI, and investigate longer-run consequences for community membership and the composition of human expertise.

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